

Daniel Scalena

📍 Italy | Netherlands ✉ scalena99@gmail.com ☎ +39 3318397458 🌐 danielsc4.it

Education

Double Doctorate (Ph.D) in Computer Science - NLP Nov 2023 – exp. Q4 2026

University of Milano - Bicocca | University of Groningen (CLCG)

Researching interpretability to improve the safety, reliability, and real-world applicability of generative models.

Mediterranean Machine Learning (M2L) Summer School Sep 2024

Milano, Italy

1-week full-time lectures and practical sessions led primarily by Google DeepMind scientists; presented my [paper](#).

M.Sc in Computer Science, ML track Oct 2021 – Oct 2023

University of Milano - Bicocca

- Final Grade: 110/110 with honors, (GPA: 29.7/30)
- Thesis: On the explainability of Large Language Models detoxification.

B.Sc in Computer Science Oct 2018 – Jul 2021

University of Milano - Bicocca

- Final Grade: 110/110 with honors, (GPA: 28.3/30)
- Thesis: Hate speech detection on social networks using state-of-the-art NLP techniques.

Experience

University of Groningen, Computational Linguistic Group Apr 2023 – Jul 2023

Research Intern

- Aligned LMs to counter hate speech narratives using RLAIIF, reaching up to 50% in safety improvements;
- Analyzed and interpreted model behavior shifts after applying post-training algorithms.

C.I.N.I. & University of Milano - Bicocca Jun 2022 - Apr 2023

Assistant Researcher

- Devised Open Relation Extraction for Italian legal texts, improving previous accuracy by 30%;
- Reduced manual legal document review using ad hoc models;
- Project funded by the *Italian Ministry of Justice* and *Inter-university Consortium for Computer Science*.

TESTUDO s.r.l. Jun 2022 - Aug 2022

ML Engineer (contractor)

- Designed and trained deep learning models to predict working hours timestamps;
- Fully automated the company's recognition system.

Main publications

🔗 **EAGER: Entropy-Aware GENEration for Adaptive Inference-Time Scaling**

Daniel Scalena, Leonidas Zotos, Elisabetta Fersini, Malvina Nissim, Ahmet Üstün

ArXiv preprint - under review, submitted to ICLR 2026

🔗 **Steering Large Language Models for Machine Translation Personalization**

Daniel Scalena*, Gabriele Sarti*, Arianna Bisazza, Malvina Nissim, Elisabetta Fersini

ArXiv preprint - under ARR review

🔗 **Multi-property steering of large language models with dynamic activation composition**

Daniel Scalena, Gabriele Sarti, Malvina Nissim

Proceedings at the Seventh BlackBoxNLP (at EMNLP 2024)

[🔗](#) **A gentle push funziona benissimo: making instructed models in Italian via contrastive activation steering**

Daniel Scalena, Gabriele Sarti, Malvina Nissim, Elisabetta Fersini

Proceedings at the Tenth Italian Conference on Computational Linguistics, CLIC-it 2024

[🔗](#) **Let the Models Respond: Interpreting Language Model Detoxification Through the Lens of Prompt Dependence**

Daniel Scalena, Gabriele Sarti, Malvina Nissim, Elisabetta Fersini

Extended abstract at the Sixth BlackboxNLP Workshop (at EMNLP 2023)

[Full list of publications](#) [🔗](#)

Scholarships and Awards

Google and Apple - BlackBoxNLP (EMNLP 2023) travel award	2023
Full-Fee M.Sc. Scholarship due to merit and financial scores	2021 - 2023
Full-Fee B.Sc. Scholarship due to merit and financial scores	2018 - 2021

Projects

Reward LM	DanielSc4/RewardLM 🔗
- Python toolkit library for Fine-Tuning and Reinforcement Learning from AI Feedback of Generative LMs; - Developed tools to interpret and compare toxicity differences between original and fine-tuned models.	

Activities

Co-organisier	Challenge the Abilities of LAnguage Models in ITALian (CALAMITA) 2024
Teaching	NLP course labs, MSc Data Science @ UniMiB (2024 - 2025)
Reviewer	EMNLP 2025, BlackBox-NLP 2025, ACL 2025, LREC-Coling 2024, NAACL 2024*, Information Processing and Management 2024 - 2025, CliC-IT 2023 - 2024, ICML Mechanistic Interpretability Workshop 2024* (*secondary reviewer)
Volunteer	CliC-IT 2021, EMNLP 2023
Misc.	European Researchers Night GroNLP collaborator 2023

Languages

Italian	Native Language
English	C1 - B-between Foreign Languages 🔗
French	B1